

Power Architecture Derivations

Outside Options, Veto Power, and Proposal Power

Working formal appendix

2026-05-10

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1 Status

This is a working derivation document, not the paper. The active manuscript `formal_model_v5.Rmd` should remain frozen until the results here are fully rederived and audited.

Protocol audit update (2026-05-10). The previously consolidated “pooling-or-delay” R1 baseline is no longer treated as verified. Its no-information delay branch used a continuation history that was not yet stated as a primitive of the extensive-form game. Pooling, delay, rejection histories, and off-path beliefs may not be imposed. They must be derived as incentive-compatible behavior under the stated protocol, or the protocol change must be explicitly approved before any proof uses it.

Feasibility audit update (2026-05-10). The C-or-R R1 characterization is superseded. A weak proposer may make an offer that is infeasible in the low state but feasible in the high state. This high-state-only branch generates separation through BF feasibility rather than through H’s preference constraints. The R1 problem is therefore C-B-R: accepted pooling, high-state-only agreement, or rejection/continuation.

Architecture reset (2026-05-11). Attempts to close the high-state-only feasibility branch exposed unresolved protocol dependence: the result depends on when feasibility is checked, whether no-consent histories reveal the state, and whether the low type of H can block before implementation. The new working decision is to abandon feasibility-based separation as the main architecture. Proposals should instead be modeled as relative institutional packages or shares that are feasible in every realized state. Screening should come from type-dependent relative participation constraints for H , not from offers that fit only one realized pie. The old C-B-R derivations remain useful as diagnostic history, but they should not be transported to the paper.

The purpose of this document is to rebuild the formal architecture from primitives after the correction that the hegemon’s majority outside option is external to the institutional pie.

1.1 How to use this document

This document is the work surface for the new formal section. It should contain the exact derivations before anything is moved back into the manuscript.

Use it in this order:

1. State the primitives and timing.
2. Derive the baseline with $\pi_H = 0$.
3. Run the outside-option stress test $\alpha = 0$ or $d_H = d_W$.
4. Prove conditional dominance and entry/nesting, if true.
5. Add extensions for $\pi_H = 1/N$ and $\pi_H > 1/N$ using bounds or simulations when the H-proposer branch has no unique payoff function.

Do not use old formulas unless they are rederived under the current architecture. In particular, any majority payoff must keep H 's outside option external to the institutional pie.

1.2 Manuscript freeze rule

Do not edit `formal_model_v5.Rmd` while this document has pending core results. The paper should receive only results that satisfy four checks:

1. the result is derived from the primitives stated here;
2. the result has a clear status in the proof ledger;
3. numerical/calibrated claims are reproduced by an R script;
4. no theorem statement depends on the old random-H-proposer architecture unless it is explicitly labeled as an extension.

Additional protocol-discipline rule: a proof may not introduce a new strategic option, voting history, tie-breaking convention, information structure, contract space, or continuation protocol as if it were already in the model. If a proof needs such an object, the result must be labeled `pending protocol decision` until the user approves the primitive and the result is rederived.

1.3 Immediate writing target

The next section to write is the baseline majority derivation under $\pi_H = 0$. It should answer:

1. What does H receive under majority when weak states control the agenda?
2. Does the result remain affine in μ ?
3. What is the weak-state payoff and entry condition?
4. Which old majority formulas survive, and which disappear because H is no longer recognized as proposer?

1.4 Equilibrium standard

The solution concept is Perfect Bayesian Equilibrium. Every formal result in this document should specify whether it characterizes:

1. a unique strategy profile;
2. a unique payoff vector despite multiple strategies at ties;
3. a payoff correspondence; or
4. a selected equilibrium under an explicit tie-breaking/refinement convention.

For each proposed equilibrium, the proof must verify:

1. **Strategies:** proposer offers, voting/acceptance rules, entry decisions, and institutional choice.

2. **Beliefs:** weak-state beliefs about θ after acceptance/rejection histories, including off-path histories when needed.
3. **Sequential rationality:** no player wants to deviate at any information set.
4. **Bayesian consistency:** beliefs are updated by Bayes' rule on path.
5. **Feasibility:** accepted proposals fit inside the realized pie in the state in which they pass.
6. **Equilibrium status:** existence, uniqueness, multiplicity, or payoff uniqueness.

At screening cutoffs, pure strategies may not be unique because proposers are indifferent between two offers. The target should be payoff uniqueness, or else an explicit tie-breaking convention. If a branch yields a payoff correspondence, it should not be used as the main theorem without selection-free bounds.

1.5 Comparative statics requirements

For every theorem, proposition, lemma, or corollary, the write-up should include the nine presentation items used by the formal-model-presentation checklist:

1. closed-form boundary;
2. comparative statics for every parameter appearing in the condition;
3. phase or region diagram for key two-dimensional parameter spaces;
4. parametric window showing the result holds on an open set;
5. margin table at the OPEC/baseline calibration;
6. verbal intuition before the formal statement;
7. worked numerical example;
8. mapping from model objects to the OPEC case;
9. proof location.

The minimum comparative statics set for the baseline is:

1. N : number of states.
2. $q = \lfloor N/2 \rfloor + 1$: majority quota.
3. r : high-state surplus.
4. α : strength of H 's outside option.
5. β : discount factor.
6. μ : belief that the high state obtains.
7. c : weak-state entry cost.
8. π_H : recognition probability, for extensions only.

The baseline proof should derive how the relevant thresholds and payoff gaps move with α , r , β , N , and c . The extension should then analyze how increasing π_H changes the bounds or calibrated payoff regions.

2 Q&A

What is the substantive puzzle?

Why would a powerful hegemon choose unanimity or consensus when that rule appears to constrain its formal power?

What mechanism should the model isolate?

Unanimity may favor the hegemon because it combines veto power with private information about

the value of the hegemon's outside option. Weak proposers must buy the hegemon's approval without observing its type.

Which sources of power must be separated?

1. Outside-option power: the hegemon has an external option $d_H(\theta) = \alpha V(\theta)$.
2. Veto or pivotality power: under unanimity, every agreement must be accepted by the hegemon.
3. Proposal power: the hegemon may control the agenda with recognition probability π_H .

What is the clean baseline?

Set $\pi_H = 0$. Weak states make proposals; the hegemon remains a privately informed veto player. This asks whether unanimity can favor the hegemon even when the hegemon has no formal agenda power.

3 Primitives

There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Let $q = \lfloor N/2 \rfloor + 1$ be the majority quota.

Nature draws a state $\theta \in \{0, 1\}$. The institutional pie is

$$V(0) = 1, \quad V(1) = r, \quad r > 1.$$

The hegemon observes θ . Weak states share prior belief $\mu = \Pr(\theta = 1)$. Let

$$V_e(\mu) = 1 + \mu(r - 1).$$

Weak states' outside option is normalized to zero. The hegemon's external outside option is

$$d_H(\theta) = \alpha V(\theta), \quad 0 \leq \alpha < 1/r.$$

The discount factor is $\beta \in (0, 1]$.

Agenda power is represented by a recognition probability

$$\pi_H \in [0, 1], \quad \pi_W = 1 - \pi_H.$$

The weak states are symmetric among themselves. Hence each weak state is recognized with unconditional probability

$$\Pr(W_i \text{ recognized}) = \frac{1 - \pi_H}{N - 1}.$$

Equivalently, conditional on the event that some weak state is recognized, each weak state is selected with probability $1/(N - 1)$. In the baseline $\pi_H = 0$, this conditional and unconditional probability coincide: $\Pr(W_i \text{ recognized}) = 1/(N - 1)$. In the neutral BF case $\pi_H = 1/N$, each player has unconditional recognition probability $1/N$. When $\pi_H > 1/N$, each weak state's unconditional

recognition probability is below $1/N$, although weak states remain symmetric conditional on weak-state recognition.

4 Architecture

The model family has three cases.

Case 1: screening baseline.

$$\pi_H = 0.$$

Only weak states are recognized. This is the main formal baseline.

Case 2: neutral BF recognition.

$$\pi_H = 1/N.$$

This restores canonical symmetric recognition. Because H is informed, the H-proposer branch is a signaling problem and may produce payoff correspondences rather than a unique payoff function.

Case 3: hegemonic agenda power.

$$\pi_H > 1/N.$$

This adds proposal power to outside-option and veto power. It should be treated as an extension after the baseline is solved.

5 Baseline Timing: $\pi_H = 0$

Working convention for the first proof pass: $\pi_H = 0$ in both bargaining rounds. This fully removes the informed-proposer signaling branch from the baseline.

1. H chooses an institutional rule $R \in \{M, U\}$.
2. Nature draws θ . H observes θ ; weak states observe only μ .
3. Weak states decide whether to enter collectively.
4. Round 1 proposer is drawn uniformly from the weak states.
5. Under majority, the weak proposer may exclude H and form a winning coalition of weak states.
6. Under unanimity, the weak proposer must obtain H 's approval and faces a screening problem.
7. If the proposal is rejected, the game proceeds to Round 2, with recognition again restricted to weak states under the baseline convention.

6 Objects To Derive

The baseline proof must derive the following objects from primitives.

6.1 Majority

Under majority, weak states can exclude H . The hegemon receives its external outside option $d_H(\theta)$; this outside option is not paid from the institutional pie.

Result M1: majority pass-branch payoffs under weak-state agenda (verified: independent PASS without reservations).

Under $\pi_H = 0$, majority generates no screening payoff. On the Round-1 low-state feasibility region

$$\mathcal{P}_M^F = \left\{ \mu \in [0, 1] : \frac{\beta(q-1)V_e(\mu)}{m} \leq 1 \right\},$$

weak states can pass a majority proposal in Round 1. If weak states enter and $\mu \in \mathcal{P}_M^F$, H 's type-contingent payoff in either bargaining round is

$$V_H^M(\theta) = \alpha V(\theta).$$

Before θ is realized, this gives

$$V_H^M(\mu) = \alpha V_e(\mu) = \alpha\{1 + \mu(r-1)\}.$$

Each weak state's expected payoff under majority is

$$V_W^M(\mu) = \frac{V_e(\mu)}{m} = \frac{1 + \mu(r-1)}{N-1}.$$

Thus the weak-state entry condition on the feasible Round-1 pass branch is

$$c \leq \frac{V_e(\mu)}{m}.$$

Equivalently, the Round-1 pass-branch majority formation set is

$$F_M^{pass} = \mathcal{P}_M^F \cap \left\{ \mu \in [0, 1] : c \leq \frac{V_e(\mu)}{m} \right\}.$$

When (M-F) holds for every $\mu \in [0, 1]$, this simplifies to

$$F_M^{pass} = \begin{cases} [0, 1], & c \leq 1/m, \\ \left[\frac{mc-1}{r-1}, 1 \right], & 1/m < c \leq r/m, \\ \emptyset, & c > r/m. \end{cases}$$

This simplification applies to the OPEC calibration below. If $\mu \notin \mathcal{P}_M^F$, the Round-1 majority pass branch is infeasible; the delayed/rejected-proposal branch remains to be derived before making a global majority-payoff statement outside $\mathcal{P}_M^F$.

Derivation.

In Round 2, a recognized weak proposer forms a minimum winning coalition of weak states. Because weak states have zero outside option and the terminal-round acceptance convention gives acceptance at indifference, the proposer can assign zero to the $q - 1$ weak coalition partners and keep the institutional pie. Since recognition is uniform over the $m = N - 1$ weak states,

$$V_{W,2}^M(\mu) = \frac{V_e(\mu)}{m}.$$

The hegemon is excluded from the majority coalition and receives the external outside option,

$$V_{H,2}^M(\theta) = d_H(\theta) = \alpha V(\theta).$$

In Round 1, a weak voter asked to support the proposal can guarantee $\beta V_{W,2}^M(\mu) = \beta V_e(\mu)/m$ by rejecting and waiting for Round 2. The weak proposer therefore offers each of $q - 1$ coalition partners the fixed transfer

$$\frac{\beta V_e(\mu)}{m}.$$

The total promised transfer is

$$\frac{\beta(q-1)V_e(\mu)}{m}.$$

Thus strict low-state feasibility requires

$$\frac{\beta(q-1)V_e(\mu)}{m} \leq 1. \tag{M-F}$$

When (M-F) holds, the proposal is feasible in both states. The proposer keeps expected payoff

$$\left(1 - \frac{\beta(q-1)}{m}\right) V_e(\mu).$$

Under symmetric random recognition and symmetric selection of coalition partners, a weak state's ex ante Round-1 payoff is

$$\begin{aligned} V_{W,1}^M(\mu) &= \frac{1}{m} \left(1 - \frac{\beta(q-1)}{m}\right) V_e(\mu) + \frac{m-1}{m} \frac{q-1}{m-1} \frac{\beta V_e(\mu)}{m} \\ &= \frac{V_e(\mu)}{m}. \end{aligned}$$

Because the majority proposal passes in Round 1, H 's Round-1 payoff is again $\alpha V(\theta)$, and its ex ante payoff is $\alpha V_e(\mu)$.

Comparative statics.

The majority payoff to H is affine in μ , increasing in α , and increasing in r whenever $\mu > 0$:

$$\frac{\partial V_H^M}{\partial \mu} = \alpha(r-1), \quad \frac{\partial V_H^M}{\partial \alpha} = V_e(\mu), \quad \frac{\partial V_H^M}{\partial r} = \alpha\mu.$$

It does not depend on N , q , or β once H 's proposal probability is set to zero. The weak-state payoff is increasing in μ and r , decreasing in $m = N - 1$, and independent of α :

$$\frac{\partial V_W^M}{\partial \mu} = \frac{r-1}{m}, \quad \frac{\partial V_W^M}{\partial r} = \frac{\mu}{m}.$$

The quota q and discount factor β affect the division between the weak proposer and weak coalition partners and determine the low-state feasibility region $\mathcal{P}_M^F$, but they wash out of the ex ante symmetric weak-state payoff wherever the Round-1 majority proposal is feasible.

For the OPEC calibration $N = 13$, $r = 1.5$, and $\beta = 0.9$, the worst-case low-state transfer requirement is

$$\frac{\beta(q-1)r}{m} = 0.675 < 1.$$

Hence (M-F) holds for every $\mu \in [0, 1]$ in the baseline calibration.

Relation to the old majority formulas.

The corrected accounting principle survives: H 's outside option is external to the institutional pie. The old random-proposer coefficients λ_M^E and κ_M^E do not survive in the baseline because they included the branch in which H is recognized as proposer. Under $\pi_H = 0$, the majority benchmark is simply

$$\lambda_{M,0}^E = \alpha, \quad \kappa_{M,0}^E = \frac{1}{N-1}.$$

The reproducible numerical check for the OPEC calibration is in `scripts/verify_baseline_majority_piH0.R`.

Independent verification on 2026-05-10 returned PASS without reservations for the pass-branch statement above, including the intersection $F_M^{pass} = \mathcal{P}_M^F \cap \{\mu : c \leq V_e(\mu)/m\}$ and the caveat that the delayed branch outside $\mathcal{P}_M^F$ is not yet derived.

6.2 Unanimity

Under unanimity, a weak proposer must make an offer acceptable to H , whose type is private. The proposer chooses among:

1. $A(\mu)$: aggressive offer accepted by the low type and rejected by the high type.
2. $C(\mu)$: conservative offer accepted by both types.
3. $R(\mu)$: deliberate rejection and continuation.

All offers must satisfy strict BF feasibility in the realized state in which they pass.

Result U2: Round-2 unanimity values under weak-state agenda (verified: independent PASS without reservations).

In the terminal round, a recognized weak proposer must secure unanimity. The non-proposing weak states have zero continuation value, so the screening choice concerns only the transfer to H .

An aggressive offer pays H 's low-state outside option α . It is accepted by the low type and rejected by the high type, giving the weak proposer

$$G_A(\mu) = (1 - \mu)(1 - \alpha).$$

A conservative offer pays H 's high-state outside option αr . It is accepted by both types and gives the weak proposer

$$G_C(\mu) = V_e(\mu) - \alpha r.$$

Both offers are feasible in the states in which they pass because $\alpha r < 1$. Therefore the representative weak-state Round-2 continuation value is

$$W_2^U(\mu) = \frac{1}{m} \max\{(1 - \mu)(1 - \alpha), V_e(\mu) - \alpha r\}.$$

The Round-2 screening cutoff is unchanged by the weak-state-agenda protocol:

$$\mu_2^* = \frac{\alpha(r - 1)}{r - \alpha}.$$

For $\mu < \mu_2^*$, the aggressive offer is strictly optimal. For $\mu > \mu_2^*$, the conservative offer is strictly optimal. At $\mu = \mu_2^*$, the weak proposer is indifferent; W_2^U is unique but the low-type H payoff depends on the tie-breaking convention.

The high type receives αr in either branch: under the aggressive branch it rejects and obtains its outside option, while under the conservative branch it accepts an offer equal to its outside option. The low type receives

$$H_{2,0}^U(\mu) = \begin{cases} \alpha, & \mu < \mu_2^*, \\ \alpha r, & \mu > \mu_2^*, \end{cases}$$

with a payoff correspondence $\{\alpha, \alpha r\}$ at the cutoff unless a tie-breaking rule is imposed. The high-type value is

$$H_{2,1}^U(\mu) = \alpha r.$$

For the OPEC calibration $N = 13$, $r = 1.5$, and $\alpha = 0.19$,

$$\mu_2^* = 0.072519.$$

The reproducible numerical check is in `scripts/verify_baseline_unanimity_R2_piH0.R`.

Independent verification on 2026-05-10 returned PASS without reservations for the cutoff, payoffs, denominator $m = N - 1$, feasibility, and tie caveat.

R1 architecture warning (verified: independent PASS without reservations).

The weak-state-agenda convention in both rounds changes the old R1 aggressive branch. Suppose an R1 weak proposer tries to make an offer accepted by the low type and rejected by the high type. If the high type rejects on path, Bayes' rule gives posterior $\mu' = 1$ in Round 2. Under Result U2, both the low and high types then receive αr in Round 2: the high type receives its outside option, and the low type is pooled into the conservative high-posterior offer. Therefore both types have the same rejection continuation, $\beta \alpha r$, after a high-posterior rejection history.

Hence a strict low-accepted/high-rejected R1 offer does not exist under the baseline tie-breaking convention in which indifferent players accept. The aggressive branch can be represented only as a knife-edge selected equilibrium in which the high type rejects while indifferent at $h_A = \beta \alpha r$. This is not payoff-distinct for H , but it can affect weak-state payoffs and therefore entry. The next proof pass must decide whether to:

1. keep $\pi_H = 0$ in both rounds and replace the old strict $A/C/R$ characterization with a pooling-or-delay characterization; or
2. change the R2 continuation protocol, which may restore strict type-dependent continuation values but reintroduces agenda-power branches.

Independent verification on 2026-05-10 returned PASS without reservations for this warning for interior beliefs $\mu \in (0, 1)$. The endpoint cases $\mu = 0$ and $\mu = 1$ are degenerate and do not change the warning.

Result U1: Round-1 unanimity with high-state-only feasibility branch (verified: independent PASS without reservations after correction).

Feasibility audit on 2026-05-10 identified a missing candidate: a weak proposer may offer transfers that exceed the low-state pie but fit inside the high-state pie. Such a branch is accepted only in the high state by feasibility, not because the low type prefers rejection. Under realized-state BF feasibility, this branch is admissible. The earlier C-or-R characterization is therefore incomplete unless the model imposes a stronger robust-feasibility primitive requiring every proposal to fit all states still possible to weak states.

This pass uses the voting protocol in the game tree: a weak proposer makes an offer, players vote accept/reject, a rejected R1 proposal leads to R2, and public responses update beliefs. The rejected-proposal branch below is not a new delay action. It is implemented by a proposal that fails for a type-independent reason, so the history carries no information about H 's type.

Given the warning above, the R1 weak proposer cannot use a strict low-accepted/high-rejected accepted agreement under the baseline tie-breaking convention. The payoff-relevant candidates are:

1. $P(\mu)$: an accepted pooling proposal that both types of H accept;
2. $B(\mu)$: a high-state-only proposal that is infeasible in the low state and feasible in the high state; or
3. $R(\mu)$: a rejected proposal that sends the game to R2 without revealing H 's type.

Let

$$g(\mu) = \max\{(1 - \mu)(1 - \alpha), V_e(\mu) - \alpha r\}, \quad W_2^U(\mu) = \frac{g(\mu)}{m}.$$

An accepted pooling proposal must give H at least its discounted high-type R2 value,

$$h_P = \beta\alpha r.$$

It must also give each non-proposing weak state its discounted R2 continuation,

$$y_P(\mu) = \beta W_2^U(\mu) = \frac{\beta g(\mu)}{m}.$$

Therefore the accepted pooling proposal is feasible in the low state iff

$$\beta\alpha r + \frac{(m-1)\beta g(\mu)}{m} \leq 1. \quad (\text{U1-F})$$

When (U1-F) holds, the proposer's pooling payoff is

$$P(\mu) = V_e(\mu) - \beta\alpha r - \frac{(m-1)\beta g(\mu)}{m}.$$

Now consider a strict separating offer A that the low type accepts and the high type rejects. If the high type rejects on path, Bayes' rule gives posterior one in R2. Under $\pi_H = 0$ in R2, posterior one induces the conservative terminal offer αr . Hence both the low type and the high type would receive αr after a high-posterior R2 rejection history. The R1 incentive constraints for a separating offer h_A are therefore

$$\text{low accepts: } h_A \geq \beta\alpha r, \quad \text{high rejects strictly: } h_A < \beta\alpha r.$$

These inequalities are incompatible. At $h_A = \beta\alpha r$, both types are indifferent, and the maintained tie-breaking convention makes both types accept. Thus the candidate separating branch collapses into pooling. There is no open belief region with a strict pure-strategy low-accepted/high-rejected R1 agreement under this protocol.

Now consider the high-state-only branch B . Let $k = m - 1$ be the number of non-proposing weak voters, and let

$$T = h + ky$$

be the total transfer promised to non-proponents, where h is the transfer to H and y is the transfer to each non-proposing weak voter. A proposal with

$$1 < T \leq r \quad (\text{B-F})$$

cannot be implemented in the low state, where the pie is 1, but can be implemented in the high state, where the pie is r . The low-state failure is then a feasibility failure, not a low-type rejection.

In the high state, H 's rejection value is its discounted R2 outside-option value, so the minimum transfer that induces high-type acceptance is

$$h_B = \beta\alpha r.$$

A non-proposing weak voter is uninformed. If she accepts the high-state-only proposal, she receives y in the high state and, in the low state, the proposal fails mechanically by feasibility and the continuation is R2 with posterior zero. If she rejects, her rejection does not itself reveal θ , so her continuation is R2 at the current belief μ . Therefore, for $\mu > 0$, her incentive constraint is

$$\mu y + (1 - \mu)\beta W_2^U(0) \geq \beta W_2^U(\mu).$$

The minimum transfer is

$$y_B(\mu) = \frac{\beta W_2^U(\mu) - (1 - \mu)\beta W_2^U(0)}{\mu} = \frac{\beta\{g(\mu) - (1 - \mu)(1 - \alpha)\}}{m\mu}. \quad (\text{B-y})$$

The minimum total promised to non-proponents is therefore

$$T_B(\mu) = h_B + ky_B(\mu) = \beta\alpha r + (m - 1)\frac{\beta\{g(\mu) - (1 - \mu)(1 - \alpha)\}}{m\mu}. \quad (\text{B-T})$$

If $1 < T_B(\mu) \leq r$, the high-state branch has an exact minimum and the proposer's payoff is

$$B(\mu) = \mu\{r - T_B(\mu)\} + (1 - \mu)\frac{\beta(1 - \alpha)}{m}. \quad (\text{B})$$

If $T_B(\mu) \leq 1$, the branch remains feasible for any $T \in (1, r]$, but the best payoff is a supremum rather than a maximum because strict low-state infeasibility requires $T > 1$. In that case the pure-strategy existence issue must be handled explicitly, for example by a discrete transfer grid or by replacing strict infeasibility with a closed protocol. In the OPEC calibration, the B branch is chosen only where $T_B(\mu) > 1$.

Protocol note. This derivation treats low-state infeasibility as a mechanical failure of the proposal, so no type-dependent H -rejection history is generated in the low state. If the final extensive form instead lets H cast an individually observed vote on an infeasible proposal before the feasibility failure, then the off-path belief after an unexpected H rejection must be stated explicitly before the B branch can be promoted to a selection-free theorem.

On the B branch, H 's expected payoff is

$$V_{H,1}^{U,B}(\mu) = (1 - \mu)\beta\alpha + \mu\beta\alpha r = \beta\alpha V_e(\mu), \quad (\text{B-H})$$

and a representative weak state's payoff is

$$V_{W,1}^{U,B}(\mu) = \frac{(1 - \mu)\beta(1 - \alpha) + \mu(r - \beta\alpha r)}{m}. \quad (\text{B-W})$$

The rejected-proposal branch is endogenous. For the main case $\alpha > 0$, the weak proposer can offer $h_R = 0$ to H . Both types of H strictly reject: the low type can obtain at least $\beta\alpha$ by rejecting

and the high type can obtain $\beta\alpha r$. Because both types reject, the public rejection history is type-independent and Bayes' rule leaves the posterior at μ . This gives the proposer the discounted representative weak-state R2 continuation

$$R(\mu) = \beta W_2^U(\mu) = \frac{\beta g(\mu)}{m}.$$

The outside-option-neutral stress test $\alpha = 0$ is a boundary case: H cannot be made to reject strictly by a nonnegative offer under the maintained acceptance tie-breaking rule. The strict-rejection implementation of R therefore uses $\alpha > 0$; the $\alpha = 0$ case must be handled separately as a limit or with an explicitly specified weak-voter rejection selection.

Therefore the pure-strategy R1 weak-proposer payoff is

$$W_{1,prop}^U(\mu) = \max \{P(\mu) \text{ if (U1-F) holds, } B(\mu) \text{ if (B-F) holds, } R(\mu)\}.$$

Thus accepted pooling is chosen only when (U1-F) holds and

$$V_e(\mu) - \beta\alpha r \geq \beta g(\mu). \quad (\text{U1-P})$$

It must also dominate the high-state-only branch B . Otherwise, the proposer chooses between B and R . At payoff ties, the proposer is indifferent; the representative weak-state payoff can remain unique while H 's payoff may depend on which branch is selected.

If accepted pooling occurs, a representative weak state's R1 payoff is

$$V_{W,1}^{U,P}(\mu) = \frac{V_e(\mu) - \beta\alpha r}{m},$$

and H 's expected R1 payoff is

$$V_{H,1}^{U,P}(\mu) = \beta\alpha r.$$

Relative to H 's expected outside option $\alpha V_e(\mu)$, the pooling rent is

$$\Delta_H^P(\mu) = \beta\alpha r - \alpha V_e(\mu) = \alpha\{\beta r - 1 - \mu(r - 1)\}.$$

This rent is positive iff

$$\mu < \frac{\beta r - 1}{r - 1},$$

provided $\beta r > 1$. Substantively, the rent comes from the low type being paid as if it might be the high type. The high type itself does not receive more than its undiscounted outside option when $\beta \leq 1$; it accepts because waiting is discounted.

If the proposer induces rejection without information, H 's expected payoff is the discounted R2 value:

$$V_{H,1}^{U,R}(\mu) = \begin{cases} \beta\alpha V_e(\mu), & \mu < \mu_2^*, \\ \beta\alpha r, & \mu > \mu_2^*, \end{cases}$$

with the same R2 tie caveat at $\mu = \mu_2^*$.

For the OPEC calibration $N = 13$, $r = 1.5$, $\alpha = 0.19$, and $\beta = 0.9$, the high-state-only branch has an exact minimum when

$$\mu > 0.232404.$$

At the accepted-pooling feasibility boundary, $T_B(\mu) = 1.126805$; at $\mu = 1$, $T_B(\mu) = 1.258875$. Thus the B branch is exact wherever it is chosen in the OPEC calibration. Condition (U1-P) holds for every $\mu \in [0, 1]$. The accepted pooling proposal is feasible for

$$\mu \leq 0.372424.$$

The pooling rent cutoff is

$$\frac{\beta r - 1}{r - 1} = 0.700000.$$

On the feasible pooling region, P strictly dominates both B and R . Once pooling becomes infeasible, B strictly dominates R . Hence the OPEC R1 regimes are

$$P \quad \text{for} \quad \mu \leq 0.372424, \quad B \quad \text{for} \quad \mu > 0.372424.$$

In the pooling region, H 's ex ante payoff is strictly above its expected outside option. At the pooling feasibility boundary, the rent is approximately 0.031120. On the high-state-only branch, however, $V_{H,1}^{U,B}(\mu) = \beta\alpha V_e(\mu) < \alpha V_e(\mu)$ for $\alpha > 0$ and $\beta < 1$. Thus H 's conditional comparison changes discontinuously when the proposer switches from accepted pooling to high-state-only agreement.

The reproducible numerical check is in `scripts/verify_unanimity_R1_C_B_R_piH0.R`.

Independent verification on 2026-05-10 first returned FAIL because the initial B derivation treated weak-voter rejection as revealing the high state. After correcting the weak-voter IC so that $y_B(\mu)$ depends on the current belief, the verifier returned PASS without reservations. The boundary case $\alpha = 0$ also remains outside the strict-rejection implementation of R and must be handled separately.

Remaining pending derivations:

1. Independent rederivation of the $\alpha = 0$ boundary case under C-B-R.

7 Protocol Closure: Feasibility and the B Branch

Status: pending protocol decision after independent verification reservations on 2026-05-11.

Re-verification note. Independent verification identified two protocol problems that must be resolved before the B branch can be promoted to a verified result.

First, if the implementation check is run even after a voter rejects, then the rejection history itself reveals whether $1 < T \leq r$ fits the realized pie. The weak-voter rejection payoff would then be state-contingent, not $\beta W_2^U(\mu)$.

Second, if consent is checked before implementation and H is one of the required pre-check voters, the low type of H has its own consent constraint. That constraint generally fails in the calibrated B region because low H would rather withhold consent and continue at the generic no-consent belief than consent to a proposal that will fail mechanically and reveal posterior zero.

Thus the text below records the candidate mechanical-feasibility closure, but the final protocol remains pending. The next proof pass must choose explicitly between at least two protocols: (i) only non-proposing weak voters consent before the implementation check and H is asked to accept only after the check reports executability; or (ii) H consents before implementation, in which case the B branch needs a low-type H consent IC and the current calibrated high- μ B region is not verified.

This section closes the protocol interpretation behind the high-state-only branch B in Result U1. It does not add a behavioral refinement. It states which public event is generated when a proposal cannot fit inside the realized pie.

7.1 Primitive Objects

The primitive BF feasibility requirement is realized-state feasibility: implemented allocations must fit inside the realized institutional pie $V(\theta)$. A proposal specifies transfers to non-proponents. In the unanimity branch, write

$$T = h + ky,$$

where h is the promised transfer to H , y is the transfer promised to each non-proposing weak voter, and $k = m - 1$. The proposal is executable in state θ only if $T \leq V(\theta)$. Because $V(0) = 1$ and $V(1) = r > 1$, any proposal satisfying

$$1 < T \leq r \tag{PC-F}$$

fits the high-state pie and does not fit the low-state pie.

To make this primitive operational, the baseline protocol includes a nonstrategic implementation check. This is not a player and it has no discretion or private objective. After a proposal is submitted for passage, the implementation check determines whether the promised total T can be executed against the realized pie $V(\theta)$. It publicly reports only one of two events:

1. F : feasibility failure, meaning $T > V(\theta)$;
2. E : executable proposal, meaning $T \leq V(\theta)$.

No strategic actor can misreport this check.

7.2 Protocol Interpretation

Mechanical feasibility failure means that a proposal that does not fit the realized pie fails mechanically at the implementation check. The public history records this event as a feasibility failure, not as an individual vote or rejection by H .

Under this protocol, a proposal satisfying (PC-F) can pass in the high state and fail in the low state without generating any off-path belief about H 's vote. In the low state, if all required voters consent, the proposal is not executable and the public history is "feasibility failure"; since $1 < T \leq r$, this public event implies posterior zero by Bayes' rule. The failed R1 proposal consumes the first bargaining round: there is no revision, no resubmission, and no redraw of an R1 proposer. The game moves to Round 2 with posterior $\mu' = 0$ and with the same baseline recognition rule $\pi_H = 0$. In the high state, if all required voters consent, the proposal is executable and the agreement passes.

The consent and implementation timing is therefore:

1. a weak proposer announces $T = h + ky$;
2. required voters choose whether to consent;
3. if any required voter withholds consent, no feasibility check is run, the R1 proposal fails by consent, and the game proceeds to R2 after a generic no-consent history;
4. if all required voters consent, the nonstrategic implementation check determines whether $T \leq V(\theta)$;
5. if $T > V(\theta)$, the only public event is F , the R1 proposal fails, and the game proceeds to R2;
6. if $T \leq V(\theta)$, the agreement is implemented.

On the equilibrium B path, event F occurs only in the low state and implemented agreement occurs only in the high state. Individual votes are not separately reported on the F history.

For the off-path generic no-consent history, no feasibility check is run and no individual H vote is reported. The continuation belief is μ . This belief is supported by a standard sequence of state-independent weak-voter trembles: if generic no-consent is generated by weak voters with probabilities that do not depend on θ , Bayes' rule leaves the posterior unchanged. This no-learning belief is used only for the deviation payoff in the weak-voter IC. It is distinct from the on-path feasibility-failure history F , where posterior zero follows directly from $1 < T \leq r$.

This is a protocol closure, not an equilibrium result. The equilibrium result uses this public feasibility event together with the acceptance constraints below.

7.3 Equilibrium Implication for Branch B

Consider an R1 weak proposer under $\pi_H = 0$, with $\mu \in (0, 1]$. On the high-state-only branch B , low-state failure is mechanical and sends the game to R2 with posterior zero. High-state agreement passes if the feasible proposal is accepted by H and by the non-proposing weak voters.

The high type of H accepts when

$$h \geq \beta\alpha r.$$

With the maintained acceptance tie-breaking convention at indifference, the minimum transfer is

$$h_B = \beta\alpha r. \tag{PC-H}$$

A non-proposing weak voter is uninformed when voting on the proposal. If she accepts and all other required voters consent, she receives y in the high state and, in the low state, receives the discounted R2 payoff following public feasibility failure and posterior zero. If she withholds consent,

no feasibility check is run, no state-revealing public event occurs, and the continuation is R2 at the current belief μ . Hence her acceptance constraint is

$$\mu y + (1 - \mu)\beta W_2^U(0) \geq \beta W_2^U(\mu). \quad (\text{PC-W})$$

For $\mu > 0$, the minimum weak-voter transfer is therefore

$$y_B(\mu) = \frac{\beta W_2^U(\mu) - (1 - \mu)\beta W_2^U(0)}{\mu} = \frac{\beta\{g(\mu) - (1 - \mu)(1 - \alpha)\}}{m\mu}. \quad (\text{PC-y})$$

The minimum promised total is

$$T_B(\mu) = \beta\alpha r + (m - 1)\frac{\beta\{g(\mu) - (1 - \mu)(1 - \alpha)\}}{m\mu}. \quad (\text{PC-T})$$

When

$$1 < T_B(\mu) \leq r, \quad (\text{PC-BF})$$

the B branch has an exact minimum and the proposer payoff is

$$B(\mu) = \mu\{r - T_B(\mu)\} + (1 - \mu)\beta W_2^U(0) = \mu\{r - T_B(\mu)\} + (1 - \mu)\frac{\beta(1 - \alpha)}{m}. \quad (\text{PC-B})$$

If $T_B(\mu) \leq 1$, strict low-state infeasibility requires raising the total promise above one. The branch then has a supremum rather than an exact maximum unless the model adds a discrete transfer grid, a weak inequality protocol for failure at the boundary, or another explicit existence convention. The OPEC calibration uses B only where $T_B(\mu) > 1$.

7.4 Sequential Rationality Check

The branch B is sequentially rational only on the parameter and belief region where all three conditions hold.

1. H 's high type accepts the high-state feasible proposal because $h_B = \beta\alpha r$, with acceptance selected at indifference. No low-type H rejection is observed on path in the low state, because the public event is mechanical feasibility failure.
2. Each non-proposing weak voter accepts when (PC-W) holds. The transfer $y_B(\mu)$ is the minimum transfer that satisfies this uninformed acceptance constraint.
3. The weak proposer chooses B only when (PC-BF) holds and $B(\mu)$ weakly dominates the accepted-pooling branch $P(\mu)$ and the no-information rejection branch $R(\mu)$. At proposer payoff ties, the payoff correspondence must be reported unless an explicit tie-break is stated.

7.5 Protocol Caveat

The analysis above is not valid under either of two alternative protocols. First, if a feasibility check is run even after a voter withholds consent, the no-consent history becomes state-revealing for high-state-only offers and the weak-voter IC must be rederived. Second, if the protocol publicly records an individual H vote before or independently of the feasibility check, then the low-state history may contain a type-dependent voting event, and the analysis requires explicit off-path beliefs and a new sequential-rationality check. Those alternative protocols should not be used without a separate modeling decision and a new verification pass.

The reproducible numerical check remains `scripts/verify_unanimity_R1_C_B_R_piH0.R`. A protocol-specific audit report is recorded in `quality_reports/2026-05-11_protocol_closure_feasibility_B_b`. The first independent verification on 2026-05-11 returned PASS without reservations. Two subsequent independent verifications returned reservations on the timing of feasibility checking, rejection, and H 's pre-check consent. The B -branch protocol is therefore pending a modeling decision before it can be verified without reservations.

8 Architecture Reset: Relative Institutional Packages

Status: adopted for reimplementaion; derivations pending.

The protocol problem above suggests a cleaner modeling route. The main baseline should not rely on a proposal that is physically feasible in one state and infeasible in another. Instead, a proposal should be an institutional package that is well-defined in every realized state. Screening then comes from the fact that the same package gives H different relative participation payoffs depending on H 's private state.

8.1 Core Decision

Replace fixed transfers $T = h + ky$ with a package

$$x \in X.$$

The package may encode a quota rule, production share, cut obligation, enforcement provision, flexibility clause, or institutional allocation. It is not a fixed monetary transfer that must fit inside a realized pie. A package is feasible if $x \in X$, and feasibility does not depend on whether $\theta = 0$ or $\theta = 1$.

A useful reduced-form representation is a one-dimensional concession index

$$y \in [0, \bar{y}],$$

where a higher y is more favorable to H and less favorable to the weak coalition. The extensive-form bargaining problem is then about which relative package y is proposed, not about whether a promised absolute payment fits inside the realized pie.

If the package is represented as shares, the primitive object is a vector

$$s = (s_H, s_{W_1}, \dots, s_{W_k}) \quad \text{with} \quad \sum_i s_i \leq 1.$$

In state θ , implemented payoffs are proportional to the realized institutional value:

$$u_i(s, \theta) = s_i V(\theta),$$

possibly plus or minus type-dependent institutional costs. The share vector is feasible in every state whenever the shares sum to at most one.

8.2 Minimal Payoff Representation

Use the following minimal representation for the first reimplementation:

$$U_H(y, \theta) = y + b_H(\theta). \tag{RP-H}$$

Here y is the concession that the weak coalition grants to H through the institutional package. It is not a fixed transfer that must fit inside the pie. It can be interpreted as a more favorable quota, more flexibility, weaker cut obligation, or better enforcement against other members.

The term $b_H(\theta)$ is the direct benefit that H receives from the agreement itself, apart from the concession y . It may capture cartel coordination, price stability, reputational value, predictability, or the value of having other members constrained by the agreement. This component is not chosen by the weak proposer once the package class is fixed; it is a state-dependent payoff consequence of being inside the agreement.

Let $o_H(\theta)$ denote H 's outside option. In a one-shot participation problem, H accepts package y in state θ iff

$$y + b_H(\theta) \geq o_H(\theta). \tag{RP-PC}$$

Thus the type- θ acceptance threshold is

$$y_\theta^* = o_H(\theta) - b_H(\theta). \tag{RP-threshold-static}$$

In the dynamic bargaining model, replace the outside option with the discounted continuation value:

$$y + b_H(\theta) \geq \beta C_H(\theta, \mu'),$$

so

$$y_\theta^*(\mu') = \beta C_H(\theta, \mu') - b_H(\theta). \tag{RP-threshold-dynamic}$$

The screening condition is a condition on net participation thresholds:

$$y_1^*(\mu') > y_0^*(\mu'). \tag{RP-S-y}$$

In the one-shot notation, this is

$$o_H(1) - b_H(1) > o_H(0) - b_H(0). \quad (\text{RP-net})$$

Substantively, the high type requires a larger concession because its outside option net of the direct benefit of agreement is higher. This allows the model to say exactly what the paper needs: weak states do not know how much they must concede to make H prefer the agreement to its alternative.

The special case $b_H(\theta) = \lambda V(\theta)$ is admissible but should not be imposed unless useful. In that case

$$y_\theta^* = o_H(\theta) - \lambda V(\theta),$$

so screening exists only if the outside option rises more, in net terms, than the direct benefit of agreement. If $o_H(\theta)$ and $b_H(\theta)$ move proportionally, the threshold may not differ by type and the screening mechanism can disappear. The general $b_H(\theta)$ formulation is therefore the safer primitive.

8.3 Screening Primitive

The central object is H 's type-dependent participation constraint. Let

$$U_H(y, \theta)$$

be H 's payoff from accepting package y in state θ , and let

$$C_H(\theta, \mu')$$

be H 's continuation value after rejection, given the posterior μ' that follows the relevant history. The participation constraint is

$$U_H(y, \theta) \geq \beta C_H(\theta, \mu').$$

The model must impose, as an explicit primitive or derived property, that the relative concession needed to satisfy H 's participation constraint differs by type. Define the type- θ threshold

$$y_\theta^* = \inf\{y \in [0, \bar{y}] : U_H(y, \theta) \geq \beta C_H(\theta, \mu')\}.$$

The intended screening environment requires

$$y_1^* > y_0^*. \quad (\text{RP-S})$$

Condition (RP-S) is the replacement for feasibility-based separation. Weak states do not know whether H requires the low relative concession y_0^* or the high relative concession y_1^* . The information problem is therefore:

weak proposers do not know how favorable the package must be for H to prefer agreement to its outside option.

This is not an assumption to hide inside a proof. The reimplemented model must state why (RP-S) follows from the primitives, or else take it as a transparent reduced-form primitive and interpret it substantively.

8.4 Equilibrium Branches Under the Reset

The R1 unanimity problem should be rederived from three candidate branches.

1. **Pooling/high-threshold package.** The weak proposer offers $y = y_1^*$. Both types of H accept. The low type receives an informational rent because $y_1^* > y_0^*$.
2. **Low-only package.** The weak proposer offers $y = y_0^*$. The low type of H accepts and the high type rejects. The rejection history is informative and the R2 posterior must be updated by Bayes' rule.
3. **No-agreement/continuation package.** The proposer offers a package that is rejected by both types or chooses a stated continuation branch, if the extensive form includes one.

There is no high-state-only feasibility branch B in this architecture. Every proposed package is feasible. Separation, when it exists, is generated by different participation thresholds, not by mechanical failure of the package.

8.5 OPEC Interpretation

This reset fits the OPEC application better than fixed transfers. OPEC members do not bargain over a dollar transfer that must fit inside an uncertain pie. They bargain over quota rules, production shares, cuts, monitoring, exceptions, and flexibility. Members may not know the future oil price, Saudi spare capacity, or Saudi Arabia's true outside option, but they can understand the relative institutional package being proposed.

In this interpretation, H is Saudi Arabia and y measures how favorable the agreement is to Saudi Arabia: a larger production share, weaker cut obligation, more flexibility, better enforcement against others, or more favorable treatment when market conditions change. Saudi Arabia's private state affects how costly a given package is relative to acting outside the agreement. When its outside option or opportunity cost is high, it requires a more favorable package to participate:

$$y_{\text{high}}^* > y_{\text{low}}^*.$$

The formal screening claim becomes: weak states know the proposed quota package, but they do not know Saudi Arabia's relative participation threshold.

8.6 Reimplementation Consequences

The next proof pass should do the following.

1. Replace the old fixed-transfer object $T = h + k\tilde{y}$ and the B -branch feasibility object with a package $y \in [0, \bar{y}]$ or a share vector s .
2. Specify $U_H(y, \theta) = y + b_H(\theta)$, weak-state payoffs, and $C_H(\theta, \mu')$.
3. Prove or explicitly assume $y_1^* > y_0^*$, equivalently $\beta C_H(1, \mu') - b_H(1) > \beta C_H(0, \mu') - b_H(0)$ for the relevant continuation posterior.
4. Recompute R2 unanimity under the relative-package protocol.
5. Recompute R1 unanimity as pooling, low-only, or continuation.

6. Recompute majority under $\pi_H = 0$, keeping H 's outside option external to the institutional pie.
7. Rebuild entry/nesting and conditional institutional comparison.
8. Rewrite scripts under new names rather than modifying the old C-B-R scripts in place.

The old C-B-R formulas are now historical diagnostics. They should not be used as theorem statements for the reimplemented model.

Result E1: weak-state entry and nesting under $\pi_H=0$ (verified: independent PASS without reservations after correction).

Under the C-B-R characterization, the representative weak state's R1 unanimity payoff is branch-dependent:

$$V_W^{U,P}(\mu) = \frac{V_e(\mu) - \beta\alpha r}{m},$$

$$V_W^{U,B}(\mu) = \frac{(1-\mu)\beta(1-\alpha) + \mu(r - \beta\alpha r)}{m},$$

and

$$V_W^{U,R}(\mu) = \frac{\beta g(\mu)}{m}.$$

The equilibrium $V_W^U(\mu)$ is the value associated with the branch that the weak proposer chooses in Result U1, and the unanimity formation set is

$$F_U = \{\mu \in [0, 1] : V_W^U(\mu) \geq c\}.$$

On the majority pass branch, the verified weak-state payoff is

$$V_W^M(\mu) = \frac{V_e(\mu)}{m}.$$

Every unanimity branch is weakly below majority. On accepted pooling,

$$V_W^M(\mu) - V_W^{U,P}(\mu) = \frac{\beta\alpha r}{m} \geq 0.$$

On high-state-only agreement,

$$V_W^M(\mu) - V_W^{U,B}(\mu) = \frac{(1-\mu)(1-\beta + \beta\alpha) + \mu\beta\alpha r}{m} \geq 0. \quad (\text{E1-B})$$

On rejection,

$$V_W^M(\mu) - V_W^{U,R}(\mu) = \frac{V_e(\mu) - \beta g(\mu)}{m} \geq 0,$$

because $g(\mu) \leq V_e(\mu)$ and $\beta \leq 1$. Hence, on $\mathcal{P}_M^F$,

$$V_W^U(\mu) \leq V_W^M(\mu),$$

and therefore

$$F_U \cap \mathcal{P}_M^F \subseteq F_M^{pass}.$$

If (M-F) holds for every $\mu \in [0, 1]$, as in the OPEC calibration, then

$$F_U \subseteq F_M^{pass}$$

for every entry cost c . In the OPEC calibration, the minimum gap $V_W^M(\mu) - V_W^U(\mu)$ remains

$$0.021375.$$

The reproducible numerical check is in `scripts/verify_unanimity_R1_C_B_R_piH0.R`.

Result C1: conditional hegemon comparison under $\pi_H=0$ (verified: independent PASS without reservations after correction).

Compare H 's unanimity payoff to the verified majority pass-branch payoff,

$$V_H^M(\mu) = \alpha V_e(\mu).$$

On accepted pooling,

$$V_H^{U,P}(\mu) = \beta \alpha r,$$

so the conditional gap is

$$D_H^P(\mu) = \beta \alpha r - \alpha V_e(\mu) = \alpha \{\beta r - 1 - \mu(r - 1)\}. \quad (\text{C1-P})$$

If $\alpha > 0$, this gap is positive iff

$$\mu < \mu_H^* \equiv \frac{\beta r - 1}{r - 1},$$

provided $\beta r > 1$.

On high-state-only agreement,

$$V_H^{U,B}(\mu) = \beta \alpha V_e(\mu),$$

so

$$D_H^B(\mu) = \beta\alpha V_e(\mu) - \alpha V_e(\mu) = -(1 - \beta)\alpha V_e(\mu) < 0 \quad (\text{C1-B})$$

for $\alpha > 0$ and $\beta < 1$. This branch is unfavorable to H relative to majority because the low state produces only delayed continuation payoff and the high state pays only the discounted high outside option.

On a no-information rejection branch, the same logic applies in the aggressive R2 region:

$$\beta\alpha V_e(\mu) - \alpha V_e(\mu) = -(1 - \beta)\alpha V_e(\mu) \leq 0,$$

with the conservative R2 case giving $\beta\alpha r - \alpha V_e(\mu)$.

For the OPEC calibration, Result U1 gives

$$P \quad \text{for} \quad \mu \leq 0.372424, \quad B \quad \text{for} \quad \mu > 0.372424.$$

The old pooling rent cutoff is $\mu_H^* = 0.7$, but it is no longer the equilibrium institutional cutoff because the game switches to B before $\mu = 0.7$. Since $0.372424 < 0.7$, H strictly prefers unanimity on the accepted-pooling region and strictly prefers majority on the high-state-only region:

$$V_H^U(\mu) = \begin{cases} 0.256500, & \mu \leq 0.372424, \\ 0.171 + 0.0855\mu, & \mu > 0.372424. \end{cases}$$

The reproducible numerical check is in `scripts/verify_unanimity_R1_C_B_R_piH0.R`.

Result I1: institutional classification on the verified majority-pass domain (verified: independent PASS without reservations after correction).

Let

$$D_H(\mu) = V_H^U(\mu) - V_H^M(\mu)$$

denote the conditional dominance gap, evaluated on the equilibrium unanimity branch and on the verified majority pass branch. Since entry is collective, the relevant formation sets are

$$F_U = \{\mu : V_W^U(\mu) \geq c\}, \quad F_M^{pass} = \mathcal{P}_M^F \cap \{\mu : V_W^M(\mu) \geq c\}.$$

On the verified majority-pass domain, the institutional classification is:

1. If $\mu \in F_U \cap F_M^{pass}$ and $D_H(\mu) > 0$, H strictly prefers unanimity.
2. If $\mu \in F_U \cap F_M^{pass}$ and $D_H(\mu) < 0$, H strictly prefers majority.
3. If $\mu \in F_U \cap F_M^{pass}$ and $D_H(\mu) = 0$, H is indifferent.
4. If $\mu \in F_M^{pass} \setminus F_U$, majority forms and unanimity does not. In the baseline payoff accounting, H 's payoff under majority is its expected outside option $\alpha V_e(\mu)$, which is also what H receives if unanimity does not form. Hence H is payoff-indifferent between majority formation and unanimity non-formation unless an explicit institution-formation tie-break is added.

5. If neither institution forms within the verified domain, H receives the outside option and is indifferent.

This is not yet a global classification outside $\mathcal{P}_M^F$. The majority delay branch outside the verified pass-feasibility region has not been derived. A global statement is justified only when (M-F) holds for every $\mu \in [0, 1]$.

For the OPEC calibration, (M-F) holds globally, so the majority-pass domain is the whole unit interval. Result E1 gives $F_U \subseteq F_M^{pass}$. Result C1 gives the calibrated comparison:

$$\begin{aligned} \mu \in F_U, \mu \leq 0.372424 &\Rightarrow H \text{ strictly chooses unanimity,} \\ \mu \in F_U, \mu > 0.372424 &\Rightarrow H \text{ strictly chooses majority,} \\ \mu \in F_M^{pass} \setminus F_U &\Rightarrow H \text{ is payoff-indifferent absent a formation tie-break.} \end{aligned}$$

The reproducible numerical check is in `scripts/verify_unanimity_R1_C_B_R_piH0.R`.

9 Stress Test: No Hegemonic Outside-Option Advantage

The current model normalizes weak outside options to zero. Under that normalization, the limiting case in which H has no stronger outside option is

$$\alpha = 0.$$

If the model is later rewritten with a common positive outside option, the same test should be written as $d_H = d_W$ before normalization.

Question to answer:

Does veto power plus private information generate screening rents when $d_H = d_W$?

Result S1: no rent when $\alpha = 0$ (verified: independent PASS without reservations).

Under the current normalization, setting $\alpha = 0$ removes the type-dependent outside option:

$$d_H(0) = d_H(1) = 0.$$

In Round 2 under unanimity,

$$G_C(\mu) = V_e(\mu).$$

The weak proposer can offer zero to H , both types accept under the tie-breaking convention, and all institutional surplus goes to weak states. The would-be aggressive/separating branch collapses into this pooling branch: when $\alpha = 0$, both types are indifferent at a zero offer, so the high type does not reject under the model's tie-breaking convention. If one writes the counterfactual high-rejection payoff as $G_A(\mu) = 1 - \mu$, then

$$G_C(\mu) - G_A(\mu) = V_e(\mu) - (1 - \mu) = \mu r \geq 0.$$

Thus zero-offer pooling weakly dominates the counterfactual separating payoff anyway. The terminal values are therefore

$$W_2^U(\mu) = \frac{V_e(\mu)}{m}, \quad H_2^U(\theta, \mu) = 0.$$

In Round 1, the pooling transfer to H is also zero:

$$h_P = \beta \alpha r = 0.$$

The proposer pays each non-proposing weak state

$$y_P(\mu) = \frac{\beta V_e(\mu)}{m}.$$

The R1 pooling proposal is feasible in the low state iff

$$\frac{(m-1)\beta V_e(\mu)}{m} \leq 1. \tag{S1-F}$$

When (S1-F) holds, the accepted pooling payoff to H is

$$V_{H,1}^{U,P}(\theta, \mu) = 0,$$

which equals H 's outside option. If (S1-F) fails or the proposer delays, the continuation payoff to H is also zero. Therefore the informational rent is identically zero:

$$\Delta_H(\theta, \mu) = V_H^U(\theta, \mu) - d_H(\theta) = 0.$$

Thus, in this baseline, pivotality and private information alone do not generate a positive rent when H 's outside option is no stronger than weak states' outside option. The rent in Result U1 comes from the interaction between pivotality and a type-dependent outside option.

The reproducible numerical check is in `scripts/verify_baseline_stress_alpha0_piH0.R`.

Independent verification on 2026-05-10 first returned a tie-breaking reservation about the counterfactual aggressive branch. After correction, a second independent verification returned PASS without reservations.

10 Historical Baseline Results: C-B-R Status Before Relative-Package Reset

This section records the historical status of the $\pi_H = 0$ benchmark before the relative-package architecture reset. After the feasibility audit, the majority pass branch and the unanimity R2 continuation remained verified under the old transfer architecture. The R1 unanimity characterization then included the high-state-only feasibility branch generated by offers that fit the high-state pie

but not the low-state pie. Entry, dominance, and classification results were rederived under the C-B-R characterization and independently verified with PASS without reservations after correction.

The 2026-05-11 architecture reset supersedes these C-B-R results for the next paper version. They are retained as diagnostic history only. The new baseline must be reimplemented with relative institutional packages and type-dependent participation thresholds.

The earlier architecture-wide adversarial audit returned PASS without reservations on 2026-05-10, but that status is superseded by the later protocol discipline correction in `quality_reports/2026-05-10_protocol_discipline_status_correction.md`.

The baseline is intentionally unfavorable to H : weak states control the agenda in both bargaining rounds. Any payoff advantage for H therefore comes from pivotality under unanimity and from the type-dependent outside option, not from proposal power.

10.1 Theorem 1: Baseline Equilibrium Architecture

Under $\pi_H = 0$ in both rounds, the currently verified pieces are as follows.

1. Under majority, on the pass-feasible branch

$$\mathcal{P}_M^F = \left\{ \mu : \frac{\beta(q-1)V_e(\mu)}{m} \leq 1 \right\},$$

weak proposers exclude H . The verified payoffs are

$$V_H^M(\mu) = \alpha V_e(\mu), \quad V_W^M(\mu) = \frac{V_e(\mu)}{m}.$$

2. Under unanimity in Round 2,

$$W_2^U(\mu) = \frac{1}{m} \max\{(1-\mu)(1-\alpha), V_e(\mu) - \alpha r\},$$

with cutoff

$$\mu_2^* = \frac{\alpha(r-1)}{r-\alpha}.$$

3. Under unanimity in Round 1, a strict low-accepted/high-rejected branch is not available under the maintained tie-breaking convention. Realized-state BF feasibility adds a high-state-only branch B , so the weak proposer solves:

$$W_{1,prop}^U(\mu) = \max \{P(\mu) \text{ if (U1-F) holds, } B(\mu) \text{ if (B-F) holds, } R(\mu)\},$$

where

$$P(\mu) = V_e(\mu) - \beta\alpha r - \frac{(m-1)\beta g(\mu)}{m},$$

$$B(\mu) = \mu\{r - T_B(\mu)\} + (1-\mu)\frac{\beta(1-\alpha)}{m},$$

when $1 < T_B(\mu) \leq r$, and

$$R(\mu) = \frac{\beta g(\mu)}{m}.$$

Accepted pooling is chosen when it is feasible and dominates both B and R .

The R branch is implemented by a type-independent rejected proposal: for $\alpha > 0$, an offer $h_R = 0$ to H is strictly rejected by both H types, so the posterior remains μ . The B branch is implemented by a proposal with total promised transfers T satisfying $1 < T \leq r$, so agreement can execute only in the high state.

10.2 Proposition 1: Hegemonic Rent in the Baseline

Status: branch-specific under C-B-R; independent PASS without reservations after correction.

When R1 accepted pooling occurs under unanimity, H 's payoff is

$$V_H^{U,P}(\mu) = \beta\alpha r.$$

Relative to the expected outside option $\alpha V_e(\mu)$, the rent is

$$\Delta_H^P(\mu) = \alpha\{\beta r - 1 - \mu(r - 1)\}.$$

Thus, for $\alpha > 0$ and $\beta r > 1$, pooling gives positive rent exactly when

$$\mu < \frac{\beta r - 1}{r - 1}.$$

The stress test $\alpha = 0$ yields no rent:

$$\Delta_H(\theta, \mu) = 0.$$

Hence pivotality and private information alone do not generate positive rent in this baseline. The positive rent comes from the interaction between pivotality and a type-dependent outside option. On the high-state-only branch, however,

$$V_H^{U,B}(\mu) = \beta\alpha V_e(\mu),$$

so the rent relative to the expected outside option is

$$\Delta_H^B(\mu) = -(1 - \beta)\alpha V_e(\mu) < 0$$

for $\alpha > 0$ and $\beta < 1$.

10.3 Proposition 2: Entry and Nesting

Status: rederived under C-B-R; independent PASS without reservations after correction.

Let

$$F_U = \{\mu : V_W^U(\mu) \geq c\}, \quad F_M^{pass} = \mathcal{P}_M^F \cap \{\mu : V_W^M(\mu) \geq c\}.$$

On the verified majority-pass branch,

$$V_W^U(\mu) \leq V_W^M(\mu).$$

Therefore

$$F_U \cap \mathcal{P}_M^F \subseteq F_M^{pass}.$$

For the OPEC calibration, (M-F) holds globally, so

$$F_U \subseteq F_M^{pass}$$

for every entry cost c . The minimum calibrated gap $V_W^M(\mu) - V_W^U(\mu)$ is 0.021375.

10.4 Proposition 3: Conditional Institutional Preference

Status: rederived under C-B-R; independent PASS without reservations after correction.

Conditional on both institutions forming and on the verified majority-pass branch, H 's comparison is governed by

$$D_H(\mu) = V_H^U(\mu) - V_H^M(\mu).$$

For the OPEC calibration,

$$V_H^U(\mu) = \begin{cases} 0.256500, & \mu \leq 0.372424, \\ 0.171 + 0.0855\mu, & \mu > 0.372424, \end{cases}$$

while

$$V_H^M(\mu) = 0.19(1 + 0.5\mu).$$

Therefore

$$D_H(\mu) > 0 \iff \mu \leq 0.372424.$$

In the OPEC calibration, conditional on both institutions forming, H strictly prefers unanimity on the accepted-pooling region and strictly prefers majority on the high-state-only region.

10.5 Corollary: Institutional Classification

Status: rederived under C-B-R; independent PASS without reservations after correction.

The conditional classification applies on the majority-pass domain. For the OPEC calibration, because (M-F) holds globally, it applies on the full unit interval:

$$\begin{aligned}
\mu \in F_U, \mu \leq 0.372424 &\Rightarrow H \text{ strictly chooses unanimity,} \\
\mu \in F_U, \mu > 0.372424 &\Rightarrow H \text{ strictly chooses majority,} \\
\mu \in F_M^{pass} \setminus F_U &\Rightarrow H \text{ is payoff-indifferent absent a formation tie-break.}
\end{aligned}$$

This classification does not say that unanimity dominates majority everywhere. It says that, in the most unfavorable agenda baseline for H , unanimity can still generate a positive informational rent, but only on the parameter/belief region where the pooling branch is feasible and pays more than H 's expected outside option.

10.6 Verification Record

The verified baseline relies on the following reproducibility and audit files:

- scripts/verify_baseline_majority_piH0.R;
- scripts/verify_baseline_unanimity_R2_piH0.R;
- scripts/verify_unanimity_R1_C_or_R_piH0.R;
- scripts/verify_unanimity_R1_C_B_R_piH0.R;
- scripts/verify_baseline_stress_alpha0_piH0.R;
- scripts/verify_entry_nesting_C_or_R_piH0.R;
- scripts/verify_conditional_dominance_C_or_R_piH0.R;
- scripts/verify_institutional_classification_C_or_R_piH0.R;
- quality_reports/2026-05-10_baseline_piH0_independent_verification.md;
- quality_reports/2026-05-10_unanimity_R1_piH0_pooling_delay_verification.md;
- quality_reports/2026-05-10_alpha0_piH0_stress_verification.md;
- quality_reports/2026-05-10_entry_nesting_piH0_verification.md;
- quality_reports/2026-05-10_conditional_dominance_piH0_verification.md;
- quality_reports/2026-05-10_institutional_classification_piH0_verification.md;
- quality_reports/2026-05-10_final_architecture_audit_piH0.md;
- quality_reports/2026-05-10_unanimity_R1_C_B_R_piH0_derivation.md;
- quality_reports/2026-05-10_unanimity_R1_C_B_R_piH0_verification.md.

11 Remaining Work

The target theorem architecture is:

1. Majority pass-branch payoff under $\pi_H = 0$ is affine in μ and contains no screening on $\mathcal{P}_M^F$; the OPEC calibration satisfies this feasibility condition globally.
2. Unanimity payoff under $\pi_H = 0$ contains a screening term generated by H 's acceptance constraints.
3. Conditional dominance: for a stated parameter region, H 's expected payoff under unanimity exceeds its payoff under majority.
4. Entry/nesting: characterize when weak-state entry under unanimity implies weak-state entry under majority.
5. Institutional classification: combine conditional dominance and entry to classify when H chooses unanimity.

12 Extension Plan: $\pi_H > 0$

After the baseline is proved, analyze agenda power separately.

For $\pi_H = 1/N$, use the canonical BF recognition probability. The H-proposer branch is an informed-proposer signaling subgame, so the extension should use selection-free bounds and calibrated sufficient conditions unless a unique equilibrium selection is explicitly imposed.

For $\pi_H > 1/N$, interpret the increase in π_H as hegemonic agenda power. This extension should not be used to establish the main screening mechanism.

13 Proof Ledger

Object	Status	Notes
pi_H architecture	Adopted	Separates proposal power from outside-option and veto power.
Baseline pi_H=0	Adopted	Main formal baseline for exact derivation.
Relative-package architecture	Adopted for reimplementation	Proposals are quota/share/institutional packages feasible in every state. Screening comes from type-dependent relative participation thresholds $y_1^* > y_0^*$, not from state-contingent feasibility failure.
Minimal H payoff primitive	Adopted for reimplementation	Use $U_H(y, \theta) = y + b_H(\theta)$, with y the concession offered by weak states, $b_H(\theta)$ the direct agreement benefit, and acceptance threshold $y_\theta^*(\mu') = \beta C_H(\theta, \mu') - b_H(\theta)$.
R2 recognition protocol	Working convention	Use pi_H=0 in both rounds for the first pass.
Majority payoff under pi_H=0	Verified: independent PASS	On $\mathcal{P}_M^F$, $V_H^M = \alpha V_e(\mu)$, $V_W^M = V_e(\mu)/(N-1)$, and $F_M^{pass} = \mathcal{P}_M^F \cap \{\mu : c \leq V_e(\mu)/m\}$. OPEC calibration satisfies (M-F) for all μ . Delay branch outside $\mathcal{P}_M^F$ is not yet derived.
Unanimity R2 under pi_H=0	Verified: independent PASS	$W_2^U(\mu) = m^{-1} \max\{(1 - \mu)(1 - \alpha), V_e(\mu) - \alpha r\}$; cutoff unchanged at $\mu_2^* = \alpha(r-1)/(r-\alpha)$.

Object	Status	Notes
Unanimity R1 low-accepted/high-rejected branch	Verified warning	With $\pi_H = 0$ in R2, the strict low-accepted/high-rejected aggressive branch collapses to a tie at $h_A = \beta\alpha r$.
Unanimity R1 C-B-R	Superseded by architecture reset	Algebra was verified after correction under the old transfer architecture, but the required protocol closure for high-state-only feasibility is unresolved. Do not transport this result to the paper.
Protocol closure: mechanical feasibility for B	Superseded / not main path	Independent verifiers identified unresolved timing issues. If feasibility is checked after rejection, no-consent can reveal the state; if H consents before implementation, low H 's consent IC generally fails in the calibrated B region. The redesign abandons feasibility-based B as the main architecture.
Stress test $\alpha=0$	Partly pending	The R2 and accepted-pooling algebra remains useful, but strict type-independent rejection by H cannot implement R when $\alpha = 0$ under acceptance in indifference.
Conditional comparison	Superseded pending rederivation	Old comparison was branch-specific under C-B-R. It must be rederived under relative institutional packages.
Institutional classification	Superseded pending rederivation	Old classification used C-B-R OPEC regimes. It must be rebuilt after the relative-package equilibrium is derived.
Entry/nesting	Superseded pending rederivation	Old nesting used C, B, and R weak-state payoffs. It must be rebuilt under relative-package payoffs.
$\pi_H=1/N$ extension	Pending	Use bounds/simulations if payoff is selection-dependent.

$\pi_H > 1/N$ extension

Pending

Treat as agenda-power
extension.
